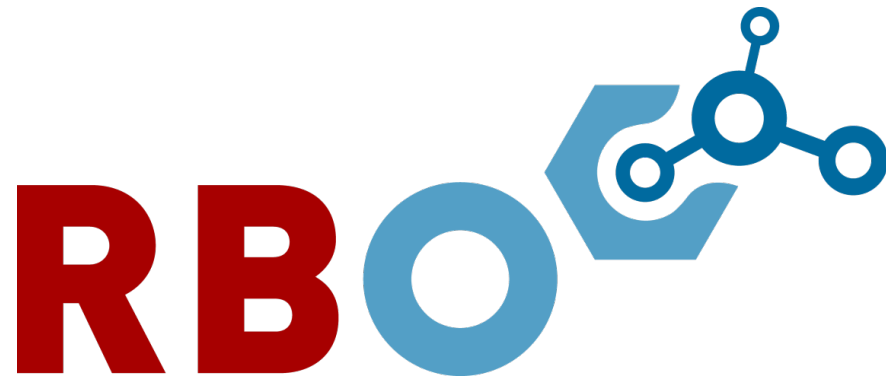
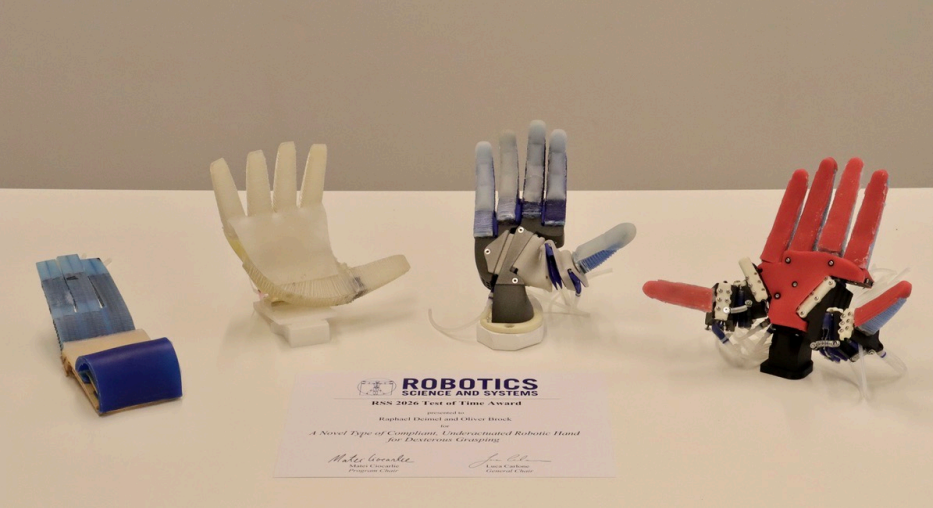




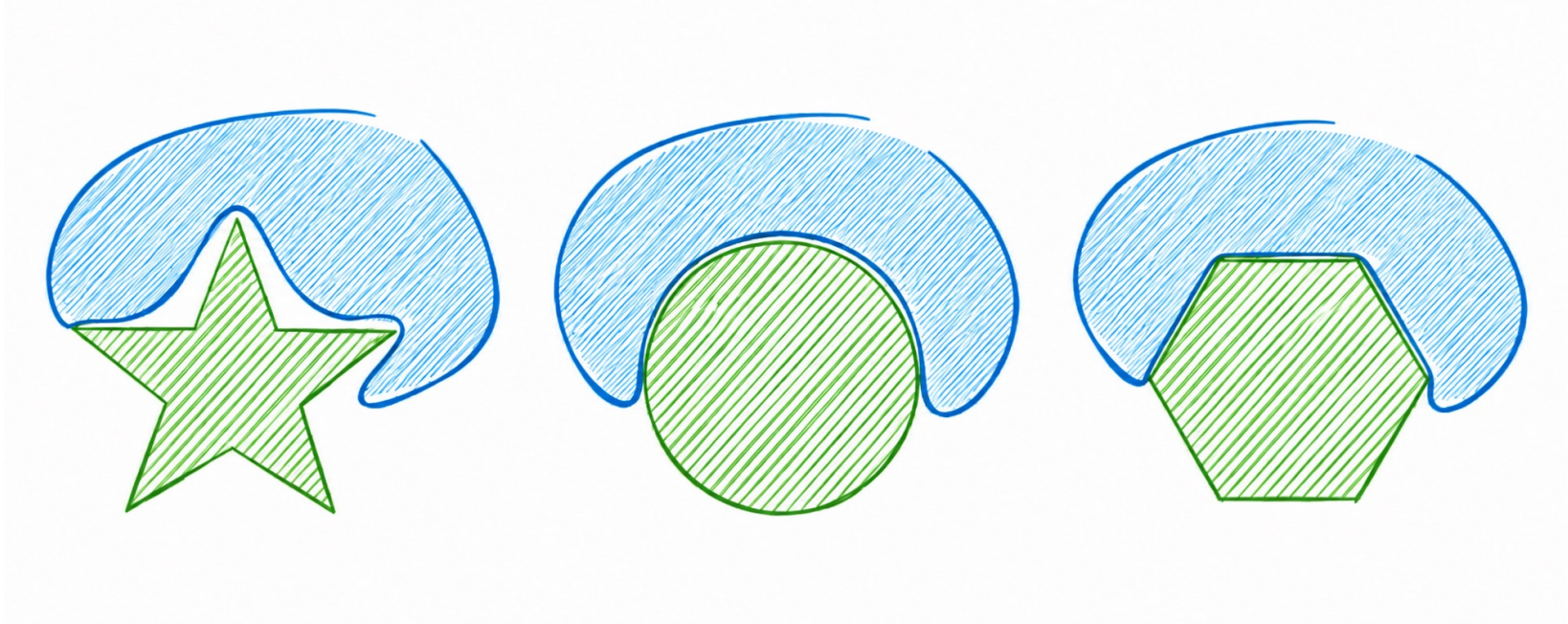
Morphological Sensing as a Bridge Between Manipulation, Perception, and Learning

Alexander Koenig

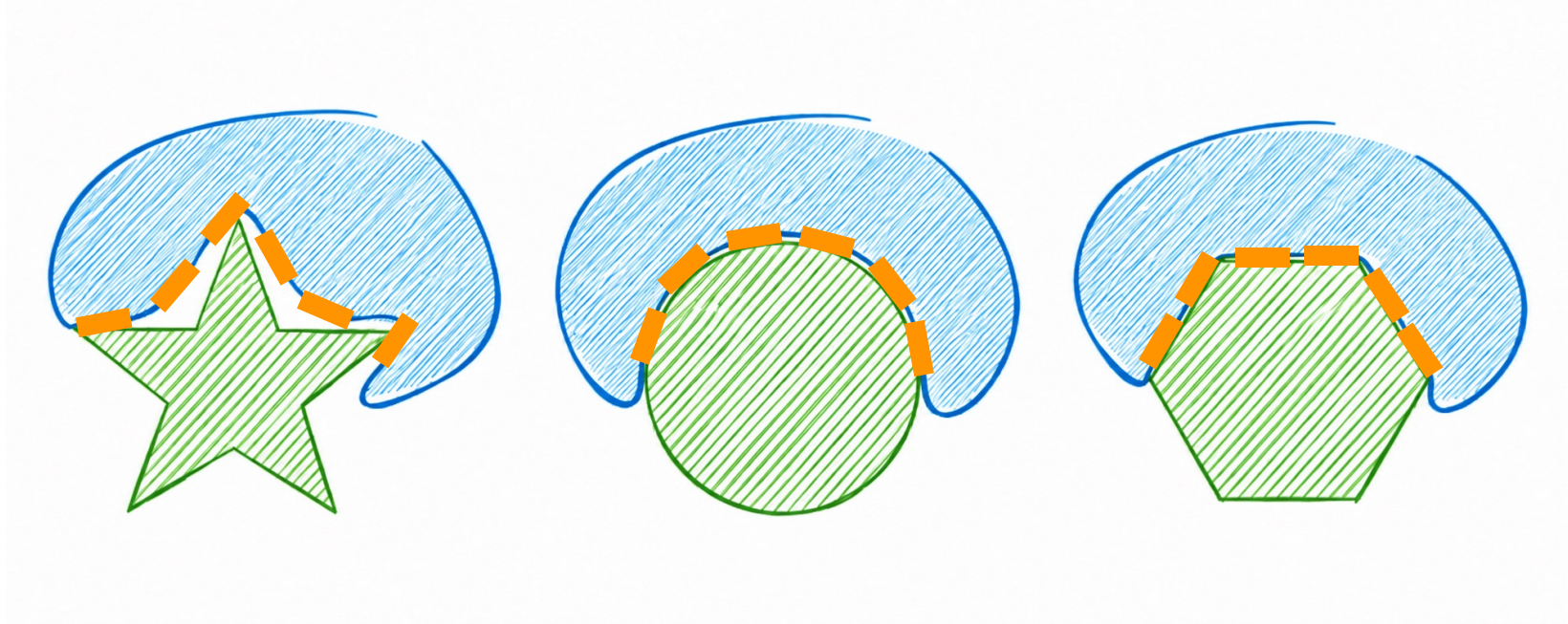
Robotics and Biology Laboratory



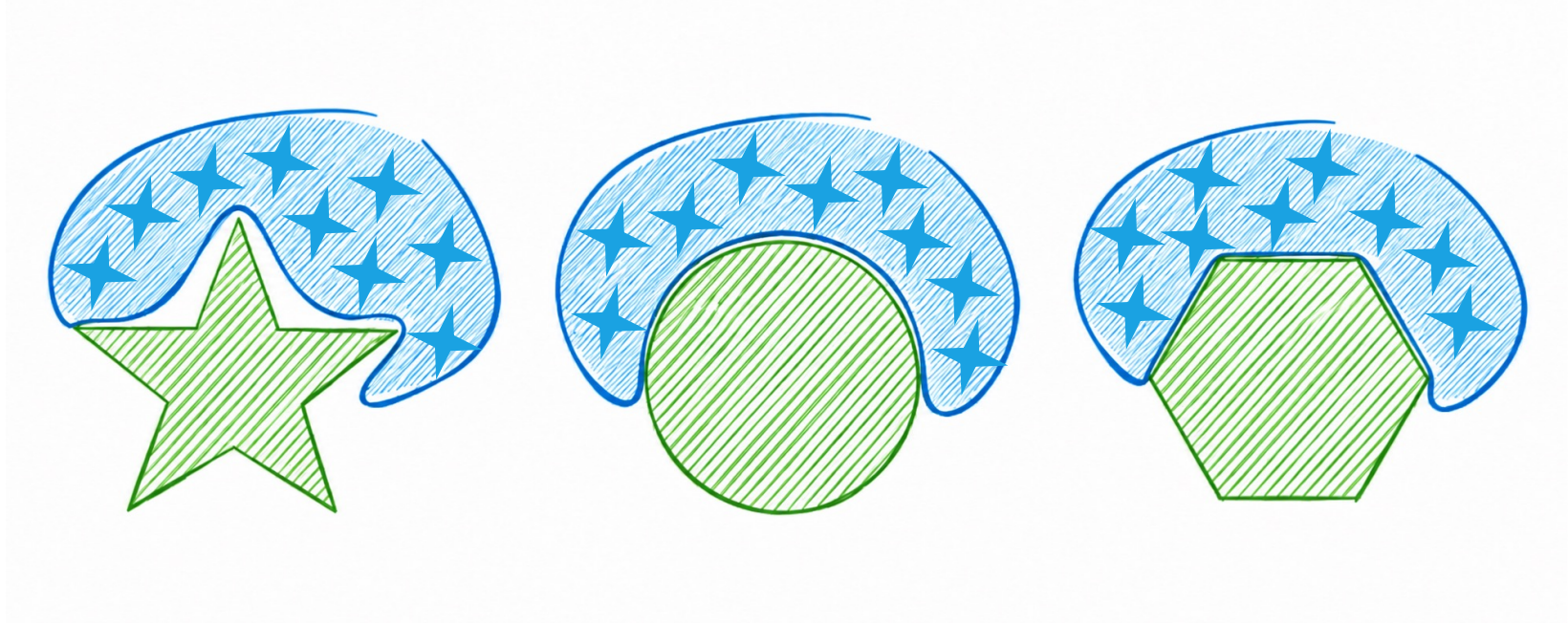
Morphology “Stores” Contact Information



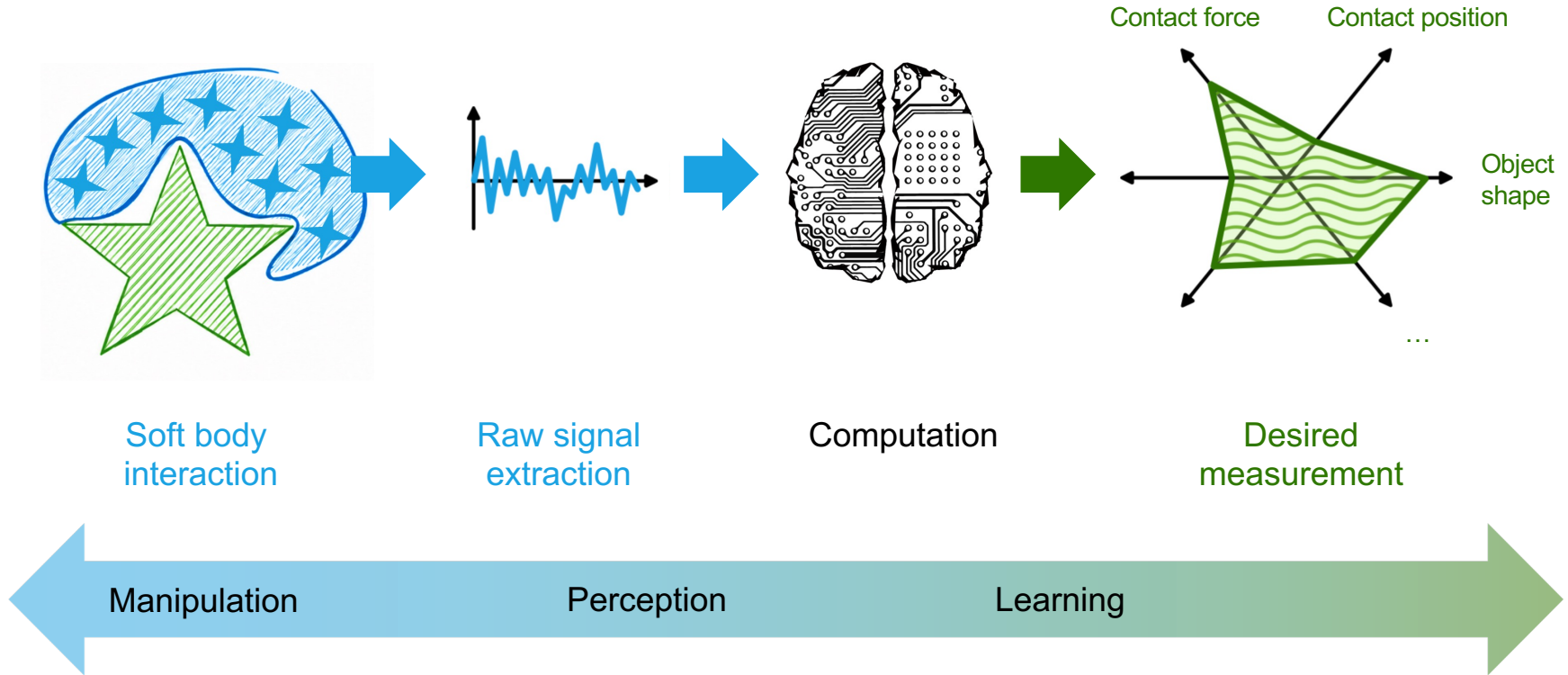
Task-Specific Tactile Sensors



Morphological Sensors

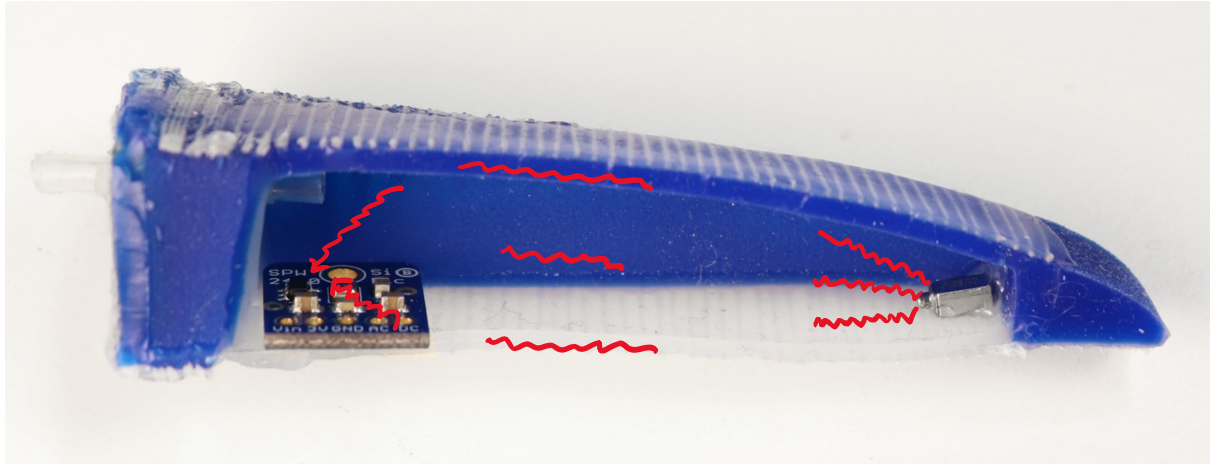


Morphological Sensing



Acoustic Sensing

Acoustic Sensing for Soft Hands



Measurements

Contact position

Contact force

Actuator inflation

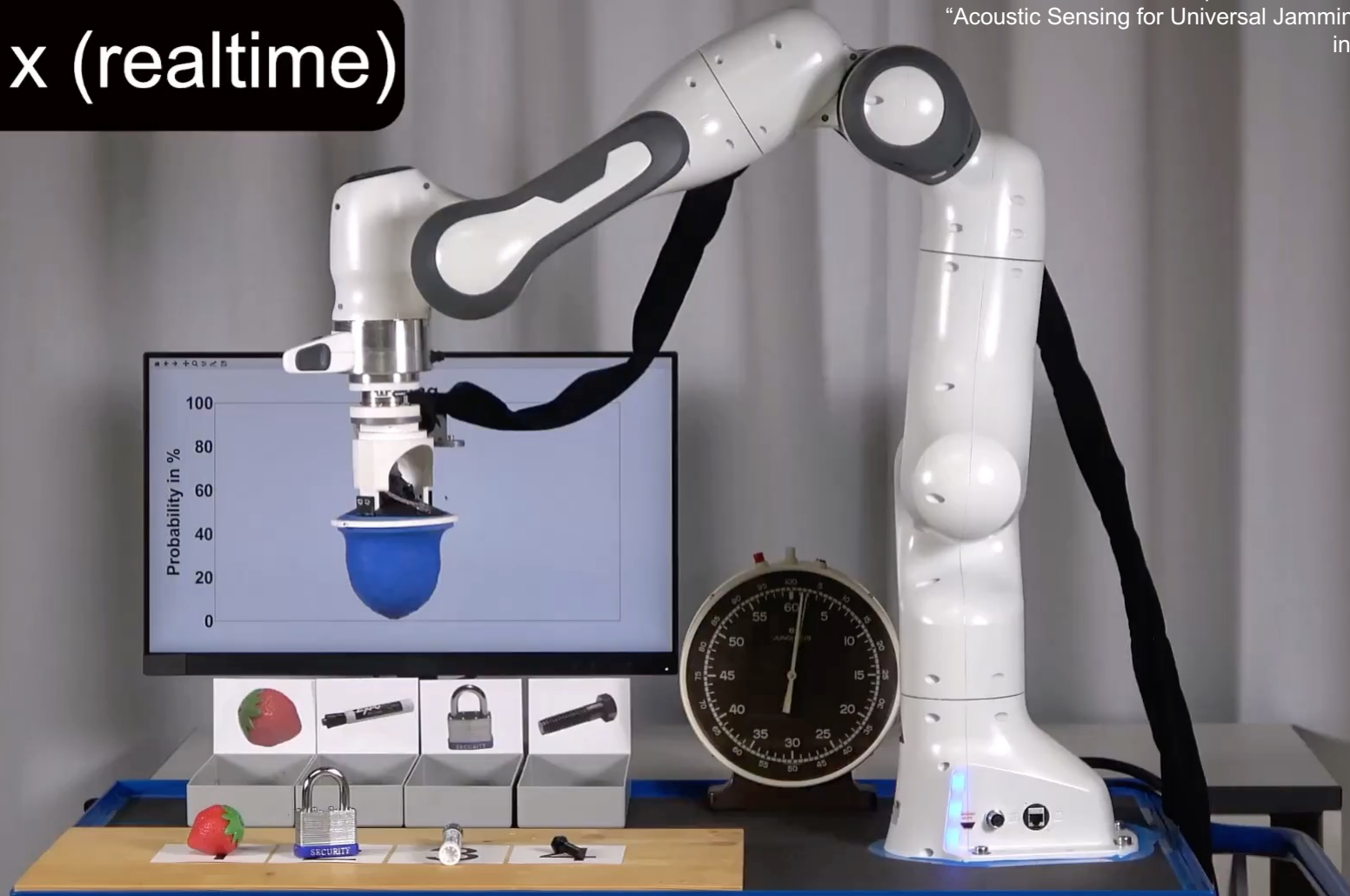
Actuator temperature

Object material

...

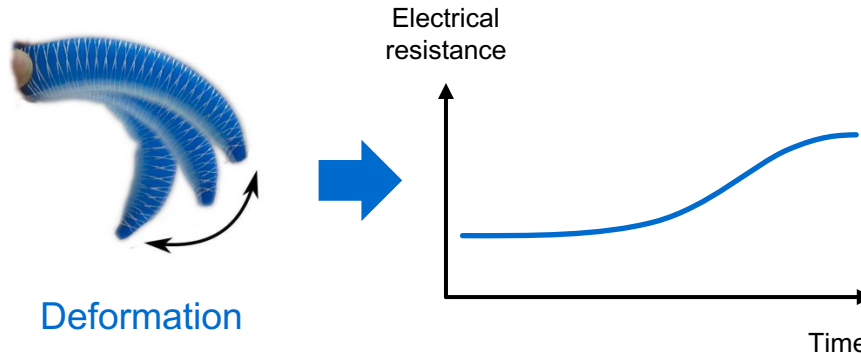
▶▶ 1x (realtime)

Weber, Wienert, Splettstößer, Koenig, Brock
"Acoustic Sensing for Universal Jamming Grippers"
in ICRA 2026



Strain Sensing

Strain Sensing for Soft Hands



Measurements

Actuator deformation

Force / torque

Contact position

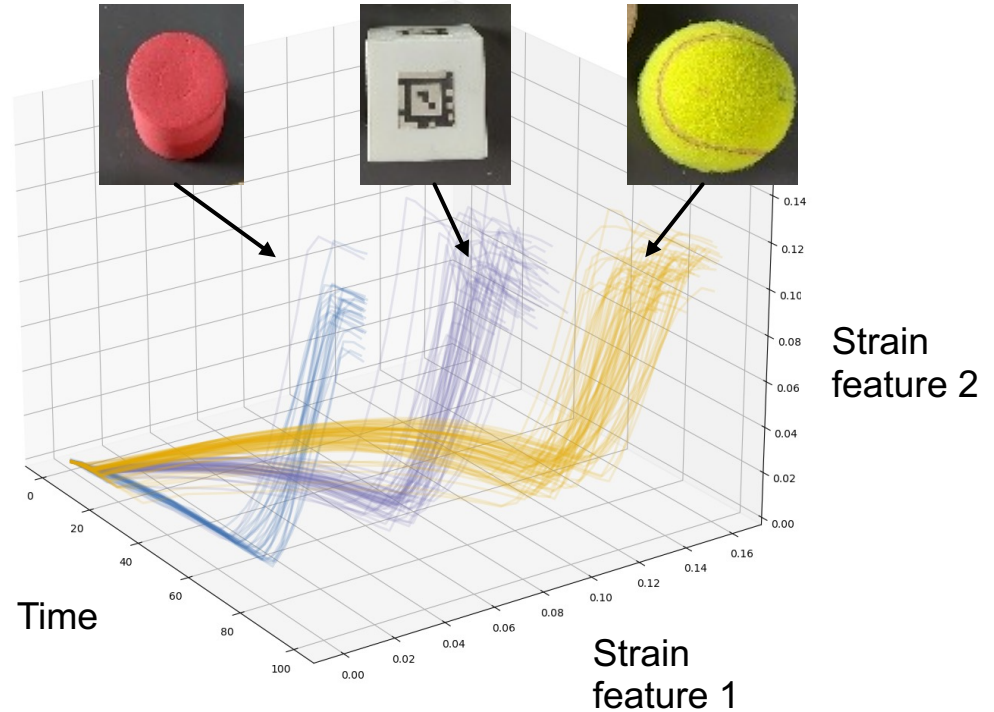
Ongoing work:
Manipulation success

...

Strain Sensing for In-Hand Manipulation



Object rotation



Morphological Sensing

